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# An On-demand Routing Scheme with QoS Provisioning for QKD Networks

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***Abstract*—As an emerging technology, Quantum Key Distribution (QKD) can provide information-theoretically secure keys for cryptographic purposes and it attracts worldwide attention. To extend its service capabilities from point-to-point communication to arbitrary remote communication parties, large-scale QKD network construction is underway. However, the dynamic and limited available resources of QKD networks remain a major obstacle to routing design. Numerous studies aim at designing routing schemes to maximize the overall network performance. While these routing schemes efficiently support end-to-end key relaying, their effectiveness diminishes when addressing diverse Quality of Service (QoS) requirements. In this paper, we propose an on-demand routing scheme with QoS provisioning to enhance end-to-end key relaying rates. Our scheme operates in two phases: an on-demand resource probing phase to dynamically update global network resource information and a routing decision phase that leverages a maximum flow model to optimize key relaying while satisfying rate-demand constraints. Simulation results on the SimQN platform demonstrate that the proposed scheme significantly improves end-to-end key relaying throughput and delay, providing effective QoS support for rate-demand QKD applications.**

***Index Terms*—Quantum Network, Quantum Key Distribution, Routing Algorithm, Quality of Service.**

## I. Introduction

Nowadays, asymmetric cryptographic algorithms based on computational security [1], such as RSA [2], are widely employed in Internet protocols to ensure the security of classical communications. However, quantum computing, based on the quantum superposition principle, enables quantum computers with exponentially enhanced computing power compared with classical computers [3], thereby threatening the security of asymmetric cryptographic schemes. Quantum Key Distribution (QKD) offers an efficient solution to address the security threats caused by quantum computing. Grounded in fundamental principles of quantum mechanics such as the no-cloning theorem and the uncertainty principle, QKD allows the distribution of information-theoretically secure keys between two communicating parties [4]. Integrating QKD with symmetric encryption algorithms can achieve unconditional communication safety according to the information theory [5], which has a promising future in the quantum era.

Currently, point-to-point QKD still faces notable limitations in terms of key generation rate, effective transmission distance, and multi-user access capability. As a result, building a scalable QKD network is essential for enabling end-to-end secure key distribution across broader areas [6]. Nowadays, plenty of testbeds are implemented in practical research, including DARPA QKD network [7], SECOQC QKD network [8], Tokyo QKD network [9], Boston QKD network [10], and Hefei metropolitan QKD network [11]. Based on these pioneering efforts, several QKD network implementation options are proposed, including trusted relay QKD networks, untrusted relay QKD networks, and optical switching QKD networks. Considering the practicality and scalability, the trusted relay is widely employed in the construction of large-scale metropolitan QKD networks in practice.

The trusted relay QKD network relies on key relaying to transmit end-to-end keys. During the process, the network finds a relaying path between the communicating parties and performs key relaying in a hop-by-hop manner [5] until the keys reach the intended node. At each hop, the preceding QKD node encrypts the end-to-end key using a point-to-point key shared with the next node and forwards it to the next hop for decryption. Thus, the performance of key relaying, including throughput and delay, is heavily influenced by the availability of the relaying path [12]. However, finding an efficient key relaying path is challenging. On the one hand, the limited key generation rate of point-to-point links increases the risk of link congestion, thereby degrading overall path performance. On the other hand, as end-to-end requests continually consume link-level keys and QKD links generate keys randomly, the frequent fluctuations of available resources severely damage the timeliness and reliability of paths. Consequently, an efficient routing algorithm is essential for QKD networks to find relaying paths with abundant resources, thus enhancing key relaying performance across the network.

Numerous efforts are devoted to designing effective routing schemes in QKD networks, which can be categorized into resource-focused routing and requirement-focused routing. Most existing studies on routing mainly focus on limited available network resources and develop routing algorithms to maximize the overall performance of networks, including throughput and delay. These schemes often involve designing link weights that account for various factors affecting link availability and using them in classical routing algorithms, like Open Shortest Path First (OSPF). For instance, the DARPA QKD network designs link weights for OSPF based on the number of residual available keys. While these schemes successfully enhance overall network performance, they lack

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mechanisms for Quality of Service (QoS) provisioning. As practical QKD networks rapidly develop, various distributed security applications utilizing QKD keys emerge, leading to diverse QoS requirements. For example, end-to-end real-time communication encrypted by quantum keys demands both low delay and high-speed key relaying to achieve real-time performance and high security. To address such differential QoS requirements, some schemes are proposed. For example, Mehic *et al.* [13] introduced GPSRQ, a QoS provisioning routing scheme designed for requests with classified delay requirements. However, both resource-focused and requirement-focused routing schemes overlook the rate-demand requirements, which are important to maintain the high security of end-to-end communications.

In this paper, we focus on the routing for end-to-end real-time communication services in QKD networks, aiming to satisfy their rate-demand QoS requirements. Thus, we propose an on-demand routing scheme with QoS provisioning for QKD networks. Our routing scheme operates in two phases: the on-demand resource probing phase and the dynamic multi-path routing phase. During the first phase, the source nodes perform on-demand resource probing by sending probe packets to nodes in the minimum vertex covering set within a single Round-Trip Time (RTT) limit. This phase ensures the collection of accurate global resource information while minimizing overhead. Based on the updated resource information, the dynamic multi-path routing phase is executed. Multi-path routing is guided by a maximum flow problem, enabling it to effectively meet the rate-demand requirements. To evaluate the performance of our routing scheme, we select three routing schemes as baselines and conduct simulations in various scenarios. Extensive results demonstrate that our scheme performs better regarding end-to-end key relaying throughput and delay, highlighting its effectiveness in QoS provisioning. The main contributions of this paper can be summarized as follows:

- To satisfy the rate-demand requirements of real-time QKD applications, we propose an on-demand routing scheme with QoS provisioning. The scheme leverages an on-demand probing mechanism to acquire the real-time status of available keys across the network and computes multi-path routes based on the required key relaying rates. Thus, our scheme not only dynamically adapts to changes in network resources but also maximizes the key relaying rate, supporting rate-sensitive applications better.
- To explore the maximum service capability of the network, we propose a network capacity model based on the maximum flow problem. Based on this model, we design an efficient multi-path routing algorithm capable of achieving the optimal solution within polynomial time complexity. The multi-path routing not only improves routing robustness but also incorporates admission control constraints to prevent network congestion.
- To evaluate the performance of our routing scheme, we conduct extensive simulations compared with existing routing schemes. Results show that the proposed routing scheme is superior to existing routing schemes in terms of throughput and delay with QoS provisioning.

The rest of this paper is organized as follows: Section II reviews some related work, while Section III introduces the system model and analyzes the main challenges in routing design. In Section IV, we present our routing scheme in detail. Section V discusses the simulation results. Finally, Section VI concludes the paper.

## II. Related Work

Currently, numerous research efforts have been made to address the challenge of finding efficient key relaying paths in QKD networks. These studies can be categorized into two main approaches: one focuses on designing routing strategies to fully use the limited network resources and enhance the overall network performance, while the other considers the various QoS requirements of different applications.

Many existing routing schemes primarily focus on the constraints of limited QKD network resources, aiming to design routing strategies accordingly that maximize network performance, while ignoring the special QoS requirements of differential applications. The DARPA QKD network, the world's first QKD network, employs the residual key quantity of each link as link weights and calculates routes using classical Dijkstra's algorithm. The SECOQC QKD network uses Dijkstra's algorithm to compute shortest path trees while performing load balancing. Tanizawa *et al.* [14] proposed a routing scheme aiming to select key relaying paths with abundant resources. Acknowledging the dynamic nature of available point-to-point keys, Yang *et al.* [15] proposed a dynamic routing scheme that comprehensively accounts for key generation rate, consumption rate, and residual key quantity. Recently, Akhtar *et al.* [16] proposed a throughput optimal routing scheme based on a maximum weight routing strategy. Although these schemes improve throughput and other performance metrics through efficient link weight designing or precise modeling in the networks, they fail to adequately address the increasingly critical QoS requirements of various applications, resulting in suboptimal performance when serving QoS-sensitive requests.

In addition to resource-focused schemes, some recent schemes take into account the varying QoS requirements of applications, trying to design routing schemes with QoS provisioning. Wang *et al.* [17] considered the differential security requirements of applications in partially trusted relay QKD networks and designed a segment-based routing scheme to meet these security needs. Chen *et al.* [18] was the first to introduce differential requirements for key quantity and key rate in the networks. Furthermore, Chen *et al.* [19], combining the concept of request priority with specific key quantity demands, proposed a routing scheme based on application priority ranking. However, the reliance on centralized controllers in these schemes limits their scalability and effectiveness in larger QKD networks. Inspired by the concept of QoS provisioning in classical networks, Mehic *et al.* proposed a

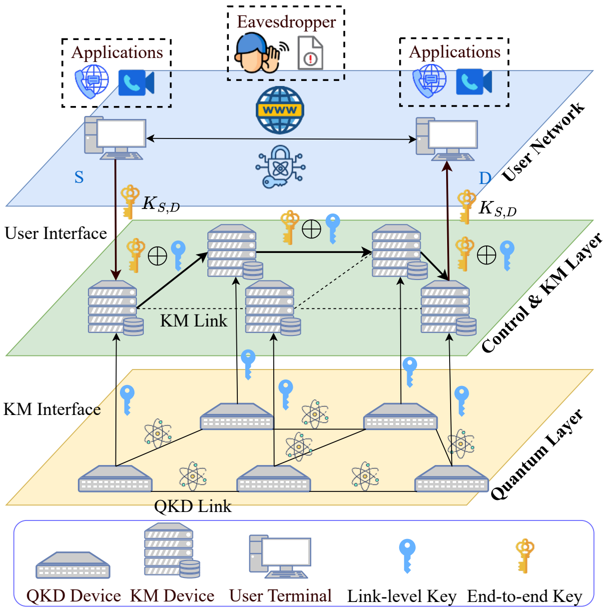



Fig. 1: The layered architecture of QKD networks

service model aimed at meeting delay-demand QoS needs in the networks. However, this scheme lacks global information over the networks, significantly reducing its routing performance of throughput. Wu *et al.* [20] introduced a distributed routing protocol based on key reservation, aiming to provide QoS service for differential delay requirements. Despite these advancements, most existing routing schemes target QoS requirements related to end-to-end key relaying delay, while routing schemes that address rate-demand QoS requirements remain underexplored.

## III. System Model And Problem Statement

In this section, we introduce the layered architecture of QKD networks employed in our scheme. Then, we present the necessary definition and the difficulty in routing design.

### A. System Model

The widely adopted layered architecture for QKD networks [21], illustrated in Fig. 1, can provide clear functional separation across its layers. From bottom to top, a QKD network is divided into the quantum layer, the key management layer, and the network control layer. The functions of each layer are as follows: In the quantum layer, the QKD devices generate unconditionally secure quantum keys, which are transmitted to the key management layer through the key management interface for further storage and use. In the key management layer, key management agents manage the life cycle of keys from the quantum layer, update the resource information of neighboring links, and perform hop-by-hop key relaying. In the network control layer, the controllers manage the control plane of the QKD network, including routing decisions, link management, and other essential functions. In the user network, the applications send their request information to the underlying network, wait for key relaying, and receive end-to-end keys via the application interface.

Based on the layered architecture, we formally model the key generation process in the quantum layer and the requests in the user network to facilitate routing design. The network is represented as an undirected graph $\mathbb{G}(\mathbb{V}, \mathbb{E})$, consisting of a set of nodes $\mathbb{V}$ and a set of links $\mathbb{E}$. A QKD node is denoted as $v_i \in \mathbb{V}$, where actual devices are implemented to generate link-level keys with neighboring nodes. A QKD link is denoted as $e_{i,j}$, where $i$ and $j$ represent the node indices at both ends of the link. The link generates new link-level keys at an average key generation rate $g_{i,j}$ and adds extra keys to the link key pool, with the current number of keys denoted as $K_{i,j}$. A request with QoS requirements in the network can be represented by a tuple denoted as $R(v_s, v_d, r_e)$, consisting of the source node $v_s$, the destination node $v_d$, the expected key relaying rate $r_e$.

In our scheme, when a QKD application in the network generates an end-to-end key relaying request $R(v_s, v_d, r_e)$, nodes located at different positions within the network play different roles and collaborate to serve the end-to-end key relaying request: The source node $v_s$ initializes and maintains end-to-end request information. It iteratively performs dynamic resource probing, collects resource information of the network, and executes routing computation until the request is finished. The intermediate nodes $v_i (i \neq s, i \neq d)$ update the key resource information of their neighboring links continuously. Upon receiving a probe packet, the intermediate node sends a feedback packet to the source node. The destination node $v_d$ receives an RTT probe packet from the source node and sends an Acknowledgment (ACK) packet back to the source node $v_s$. This feedback is used to limit the probe time and ensure timely routing decisions.

### B. Problem Statement

The rate-demand QoS requirements in QKD networks, generated by applications including end-to-end real-time communication encrypted with quantum keys, present significant challenges to routing design. From the QKD application perspective, high-speed end-to-end key delivery is essential to support symmetric cryptographic algorithms and maintain continuous, high-level security. However, current routing approaches, which make decisions on a per-packet basis, lack awareness of the overall request requirements and fail to meet the needs of continuous rate-demand requests. To address this, an on-demand routing scheme is necessary. From the QKD network resource perspective, the dynamic and limited nature of network resources introduces additional complexity. These characteristics reduce the effectiveness of existing routing schemes and increase the likelihood of link congestion. Addressing these issues requires a refined model of network resources to guide more effective routing decisions.

Motivated by these ideas, we propose a practical on-demand routing scheme with QoS provisioning for QKD networks.

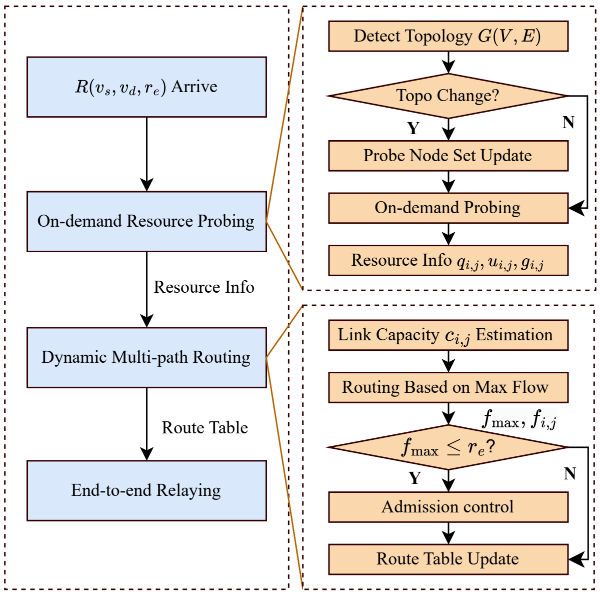



Fig. 2: The process of end-to-end key relaying and routing.

First, through on-demand probing, our scheme collects essential resource information to support informed and efficient routing decisions. Then, a maximum flow model is introduced to guide the routing process, offering a relatively accurate quantification of network service capabilities. By integrating these two phases, our scheme effectively adapts to dynamic and limited network resources, delivering enhanced performance in QoS provisioning.

## IV. Routing Scheme

In this section, we introduce the on-demand routing scheme with QoS provisioning for QKD networks, which consists of the on-demand resource probing phase and the dynamic multi-path routing phase.

### A. Overview

In this section, we discuss the overview of our on-demand routing scheme with QoS provisioning for QKD networks. As Fig. 2 shows, it can be divided into the following two phases:

**On-Demand Resource Probing Phase**: When a new request arrives or an existing request requires a route update, the source node performs on-demand resource probing. First, the source node checks topology changes in the QKD network through the Link State Advertisement (LSA) mechanism in OSPF. If topology changes are detected, such as the addition of new nodes, the source node updates the probing node set by solving a minimum vertex covering problem. Then, the source node sends probe packets to nodes in the set and collects resource information through their feedback. To achieve a timely probe, the source node only accepts feedback packets in a single RTT. The real-time resource information probed across the network, including the average key generation rate and residual key number, is then used for further decision-making. This phase enables our scheme to dynamically adapt to changes in network resources and supports efficient routing in the next phase. Details of this phase are discussed in Section IV-B.

**Dynamic Multi-path Routing Phase**: Based on the resource information obtained in the previous phase, the source node estimates the capacities of links within the detected topology. Then, the source node formalizes and solves a maximum flow problem to find the capacity of the current network and meet the QoS requirements for high end-to-end key relaying as much as possible. Afterward, a routing table is established based on the solution, guiding the end-to-end key relaying process. Additionally, admission control is applied to avoid requests exceeding the network's capacity. Details of this phase are covered in Section IV-C.

### B. On-Demand Resource Probing

In this section, we present the on-demand resource probing phase of our scheme. This phase operates in two main steps: probe node set update and on-demand probing. The probe node set update step is triggered upon topology changes in the QKD network. The source node computes the nearly optimal set of probe nodes, which ensures efficient coverage of the network information while minimizing the signaling cost. Once a request arrives, source node $v_s$ initiates resource probing, leveraging the updated probe node set. This process enables our scheme to quickly gather real-time resource information to make informed routing decisions.

*1) Probe Node Set Update:* To address the challenges caused by dynamically changing QKD network resources, an effective way is to introduce an on-demand probing mechanism, thereby updating resource information over the network. Despite achieving global resource probing, flood probing is not a suitable choice for this scenario. On the one hand, it leads to high signaling overhead. On the other hand, it struggles to achieve timely probes. Given that the key management module on nodes can continuously monitor the status of neighboring QKD links, it is sufficient to probe a small subset of nodes rather than the entire network. We model the problem as a minimum vertex covering problem, where the goal is to probe the smallest subset of nodes to collect resource information for all links in the network.

We use $x_i$ to denote whether node $v_i$ is included in the minimum vertex covering set $\mathbb{C}$. The problem can be expressed with the following optimization objective Eq. (1):

$$\min_{x_i} \quad \sum_{v_i \in \mathbb{V}} x_i \tag{1}$$

subject to:

$$x_i + x_j \geq 1, \quad \forall e_{i,j} \in \mathbb{E}, \tag{1a}$$
$$x_i = 1, \quad \text{if } v_i \text{ is chosen as a probe node}, \tag{1b}$$
$$x_i \in \{0, 1\}, \quad v_i \in \mathbb{V}. \tag{1c}$$

The objective in Eq. (1) aims to minimize the size of the vertex covering set. The constraint in Eq. (1a) ensures that the

**Algorithm 1:** Minimum Vertex Covering Algorithm Based on Local Search Optimization

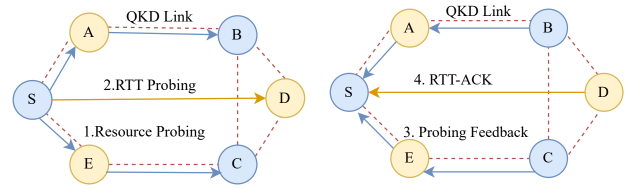
**Input:** Undirected graph $\mathbb{G} = (\mathbb{V}, \mathbb{E})$;
**Output:** Approximate minimum vertex covering set $\mathbb{C}$;

***Initialization:***
$\mathbb{C} \leftarrow \varnothing$;
$\mathbb{E}' \leftarrow E$;
***Phase 1: Approximate Solution***
**while** $\mathbb{E}' \neq \varnothing$ **do**
  Select an arbitrary edge $e_{i,j} \in \mathbb{E}'$;
  $\mathbb{C} \leftarrow \mathbb{C} \cup \{v_i, v_j\}$;
  **for** *each* $e_{m,n} \in \mathbb{E}'$ **do**
    **if** $v_i \in e_{m,n}$ *or* $v_j \in e_{m,n}$ **then**
      $\mathbb{E}' \leftarrow \mathbb{E}' \setminus \{e_{m,n}\}$;
    **end**
  **end**
**end**
***Phase 2: Local Search Optimization***
$improved \leftarrow$ true;
**while** $improved$ **do**
  $improved \leftarrow$ false;
  **for** *each* $v_i \in \mathbb{C}$ **do**
    $\mathbb{C}' \leftarrow \mathbb{C} \setminus \{v_i\}$;
    **if** $\mathbb{C}'$ *is a valid vertex cover* **then**
      $\mathbb{C} \leftarrow \mathbb{C}'$;
      $improved \leftarrow$ true;
      **Break**;
    **end**
  **end**
**end**
**return** $\mathbb{C}$;

selected nodes in the set cover all the links. Meanwhile, the constraints in Eq. (1b) and Eq. (1c) define the state of each node, indicating whether it is included in or excluded from the set. We use the solution of the optimization problem to guide the on-demand probing process.

As illustrated in Alg. 1, we propose a heuristic algorithm for solving the minimum vertex covering problem using local search optimization, as the problem is NP-hard. Initially, a random algorithm is utilized to compute an approximate solution for the minimum vertex covering problem. The size of the approximate set is guaranteed to be no more than twice the size of the minimum vertex covering set, and it can be computed in a polynomial time $O(|\mathbb{E}|^2)$. Next, we adopt local search to further optimize the solution within a polynomial time cost $O(|\mathbb{E}| \cdot |\mathbb{V}|)$. Given that the QKD network topology remains relatively stable, the computation of the probe node set can be performed offline. When changes in the network topology are detected, such as the addition of a new node or the failure of a node, the probe node set is updated accordingly.

*2) On-Demand Probing:* The process of on-demand probing is illustrated in Fig. 3. When a request arrives at the source node $v_s$, it sends probe packets to the nodes in the




(a) Perform probing. (b) Obtain feedback.

Fig. 3: The process of on-demand probing.

probe set, including $v_b$ and $v_c$, and waits for feedback. At the same time, the source node sends an RTT probe packet to the destination node $v_d$. Upon receiving the RTT probe packet, the destination node replies with an ACK packet to the source node. Any probing feedback received within a single RTT is considered valid, providing resource information for neighboring links. For example, feedback from node $v_b$ includes resource information about link $e_{a,b}$, $e_{b,c}$ and $e_{b,d}$. The on-demand probing, constrained by a single RTT, ensures the acquisition of extensive QKD network resource information within a limited time frame. This approach achieves high information accuracy and low probing delay, facilitating efficient routing decisions.

### *C. Dynamic Multi-path Routing*

In this section, we present the routing decision phase of our scheme. This phase consists of two main steps: link capacity estimation and routing based on maximum flow. In the link capacity estimation, the source node computes the link capacity for the current decision interval, designed to offer precise link capacity limits for the later maximum flow solution. After that, the source node models a maximum flow problem, which reflects the available resource status. The solution is employed to guide the routing decisions. If the network cannot fully accommodate a request, the admission control is triggered to mitigate link congestion.

*1) Link Capacity Estimation:* First, to provide more accurate relay decision support for end-to-end requests, we build a link available key quantity model based on the model introduced by Yang *et al.* [15]. The available key quantity of link $e_{i,j}$ at time $t$ is described by the following integral equation, Eq. (2):

$$q(t) = q_{\text{pro},i,j} + \int_0^t \left( g_{i,j} - \sum_{k=1}^{K} u_{k,i,j} \cdot \mathbb{I}_{[0,T_k]}(t') \right) dt', \quad (2)$$

where $q_{\text{pro},i,j}$ represents the probed number of residual available quantum keys in the key pool, $g_{i,j}$ denotes the probed average link key generation rate, $u_{k,i,j}$ is the averaged quantum key consumption rate of the request $r_k$ on the link $e_{i,j}$, $t_{\text{pro},i,j}$ is the key information probing time, and the indicator function $\mathbb{I}_{[0,T_k]}$ indicates the request to consume key resources within the time interval $[0, T_k]$.

Based on the model presented in Eq. (2), we can derive the maximum link capacity in terms of service rate within the time interval $[0, T]$, as shown in the following equation Eq. (3):

$$c_{i,j} = \frac{q_{\text{pro},i,j}}{T} + g_{i,j} - \sum_{k=1}^{K} u_{k,i,j}, \tag{3}$$

where $\sum_{k=1}^{K} u_{k,i,j}$ represents the total average key consumption rate for all requests on the link. However, directly exchanging information about $u_{k,i,j}$ for all requests imposes a significant communication overhead. To address this, we compute it by subtracting the expected key consumption rate $u_{\text{past},i,j}$ of the current request from the probed average link key consumption rate $u_{i,j}$. This approximation reduces overhead while maintaining sufficient accuracy for routing decisions.

*2) Routing Based on Maximum Flow:* To provide high key relaying rates that meet rate-demand requirements, we model the problem as a maximum flow problem within a given decision cycle $T$. By maximizing the theoretical throughput of end-to-end requests, our scheme can aggregate resources along multiple paths, thereby satisfying the rate-demand requests. At the end of each time interval, the source node performs a new round of on-demand resource probing and dynamic multi-path routing. This iterative process enables our scheme to maintain high key relaying rates for requests that persist over extended periods.

We define $c_{i,j}$ as the capacity of a link in terms of key relaying rates, which is computed by Eq. (3), $f_{i,j}$ as the key relaying traffic along the directed link $e_{i,j}$, $v_s$ as the source node, and $v_d$ as the destination node. The problem can be expressed with the following optimization objective Eq. (4):

$$\max_{f_{i,j}} \quad \sum_{e_{s,j} \in \mathbb{E}} f_{s,j} \tag{4}$$

subject to:

$$0 \le f_{i,j} \le c_{i,j}, \forall e_{i,j} \in \mathbb{E}, \tag{4a}$$

$$\sum_{e_{i,j} \in \mathbb{E}} f_{i,j} = \sum_{e_{j,m} \in \mathbb{E}} f_{j,m}, \forall v_j \notin \{v_s, v_d\} \tag{4b}$$

$$\sum_{e_{s,j} \in \mathbb{E}} f_{s,j} = \sum_{e_{i,d} \in \mathbb{E}} f_{i,d}. \tag{4c}$$

The objective of Eq. (4) is to maximize the flow, ensuring that the rate-demand requirements are met within the short time interval. Constraint Eq. (4a) enforces the capacity limits of the network flow, while constraint Eq. (4b) ensures flow conservation. Since the probed QKD network resource information is represented as an undirected graph $\mathbb{G}(\mathbb{V}, \mathbb{E})$, each undirected edge $e_{i,j}$ must be transformed into two directed edges with opposite directions and equal capacities during the solution process.

As outlined in Alg. 2, we employ the classic maximum flow solution algorithm, the Ford-Fulkerson method, to solve the maximum flow problem within a time complexity $O(|E| \cdot f_{max})$. First, we iteratively find augmenting paths from the source $v_s$ to the destination $v_d$ in the residual network. For each augmenting path, the flow is adjusted by the minimum residual capacity along the path, and the residual capacities of the edges are updated accordingly. This process repeats until no more augmenting paths can be found. Next, the maximum flow value is computed as the sum of the flows on all edges originating from the source node. Furthermore, to reduce link congestion, we impose a simple admission control. This mechanism evaluates whether the maximum flow can satisfy the request's demand; if not, only the portion of the request that is less than the maximum flow is served. Finally, we compute the routing table based on the flow distribution.

**Algorithm 2:** Multi-path Routing Based on Maximum Flow

**Input:** Residual network $\mathbb{G} = (\mathbb{V}, \mathbb{E}, \mathbb{C})$, Request $R(v_s, v_d, r_e)$;
**Output:** Routing table $\mathbb{T}$, access flow value $f_{acc}$;

***Initialization:***
$f_{i,j} \leftarrow 0, \quad \forall e_{i,j} \in \mathbb{E}$;
***Phase 1: Augmenting Path Search***
**while** *augmenting path $P$ from $v_s$ to $v_d$ exists* **do**
  $c_{min} \leftarrow \min(c_{i,j}), \quad \forall e_{i,j} \in P$;
  **for** *each edge $e_{i,j} \in P$* **do**
    **if** $e_{i,j} \in \mathbb{E}$ **then**
      $f_{i,j} \leftarrow f_{i,j} + c_{min}$;
    **end**
    **else**
      $f_{i,j} \leftarrow f_{i,j} - c_{min}$;
    **end**
  **end**
  Update the capacities of edges along path $P$;
**end**
***Phase 2: Compute Maximum Flow***
$f_{max} \leftarrow \sum_{e_{s,j} \in \mathbb{E}} f_{s,j}, \quad \forall e_{s,j} \in \mathbb{E}$;
**if** $f_{max} < r_e$ **then**
  $f_{acc} \leftarrow f_{max}$;
**end**
**else**
  $f_{acc} \leftarrow r_e$;
**end**
***Phase 3: Routing Table Construction***
Generate the routing table $\mathbb{T}$ based on $f_{i,j}$;
**return** $\mathbb{T}$ and $f_{acc}$;

## V. Simulation

To evaluate the performance of our routing scheme, we conduct simulations using a random topology generated by the Waxman model. We implement the Shortest Path First (SPF) scheme, the GPSRQ scheme [13], and the SKA scheme [15] as baselines. The whole simulations are carried out in the SimQN platform [22]. We utilize the BB84 protocol in the simulator to imitate the key provisioning capability of actual QKD devices, thus the generation key rate is set between 10Kbps and 20Kbps. The simulation results demonstrate that our scheme delivers high-speed, low-delay end-to-end key relaying services for applications with rate-demand QoS requirements. Even in scenarios when multiple requests compete for link-level keys, our dynamic multi-path routing based on maximum flow assists in keeping high end-to-end key relaying throughput and low delay performance.

### A. Simulation Setup

**Simulation Environment.** The software and hardware environment supporting the simulations are as follows: We use an Intel Core i5-13400, with 32GB of memory. The operating system is Ubuntu 22.04, with essential software dependencies including Python 3.10, SimQN 0.1.5, and Networkx 3.2.1.

**Network Topology Generation.** To create a simulation environment that closely approximates practical QKD networks, we construct a QKD backbone network simulation topology consisting of $50$ nodes, within a $500km \cdot 500km$ area. The nodes and links in the topology are randomly generated by the Waxman model, which prioritizes the formation of QKD links between geographically closer nodes. We assume that classical networks enable information exchange between any two QKD nodes. The probability of the existence of direct quantum channels between two nodes is given by Eq. (2):

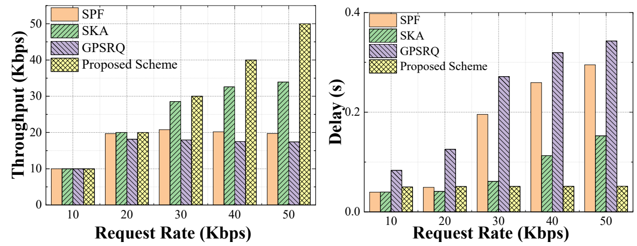
$$P_{i,j} = \alpha e^{-\text{dis}(v_i, v_j)/\beta L}, \tag{5}$$

where $\text{dis}(v_i, v_j)$ is the Euclidean distance between node $v_i$ and $v_j$, $L$ is the largest distance between any two nodes, $\alpha$ and $\beta$ are adjustable parameters influencing network connectivity, set to 0.5 and 0.3.

**Default Parameters.** The average key generation rate for each link is set between 10Kbps and 20Kbps, and a 20Mbit key pool is assigned to each link to cache unused point-to-point keys. The initial available keys for each link are set between 0.5 and 1Mbit. Random requests are generated between any two nodes in the simulation topology, which means that upper QKD applications audio between any two nodes in the QKD backbone network can generate end-to-end key relaying requests with rate-demand requirements. The simulation lasts 100s. $T$ in our scheme is set to 10s.

**Comparison Schemes.** We select three typical schemes as baselines: the SPF scheme, the GPSRQ scheme, and the SKA scheme. First, the SPF scheme is chosen to evaluate the throughput benefits that dynamic multi-path routing brings to our proposed scheme. Additionally, the main reason for selecting the GPSRQ scheme is to verify that, compared to other schemes taking QoS requirements into account, our routing scheme can provide more efficient key relaying services in satisfying requests with rate-demand requirements. Furthermore, the SKA scheme is chosen for comparison with typical routing schemes that focus on fully using the network resources and maximizing the overall network performance.

**Evaluation Metrics.** To evaluate the end-to-end key relaying QoS performance of different schemes, we select throughput and delay as the main metrics. Throughput is defined as the total amount of end-to-end keys successfully relayed per time slot. This metric reflects the maximum potential performance of the routing schemes. Delay is defined as the average time required for all successfully relayed keys. It measures the adaptability of different schemes to dynamically changing QKD network resources.




(a) Throughput for single request (b) Delay for single request

Fig. 4: Performance comparison results for a single request.

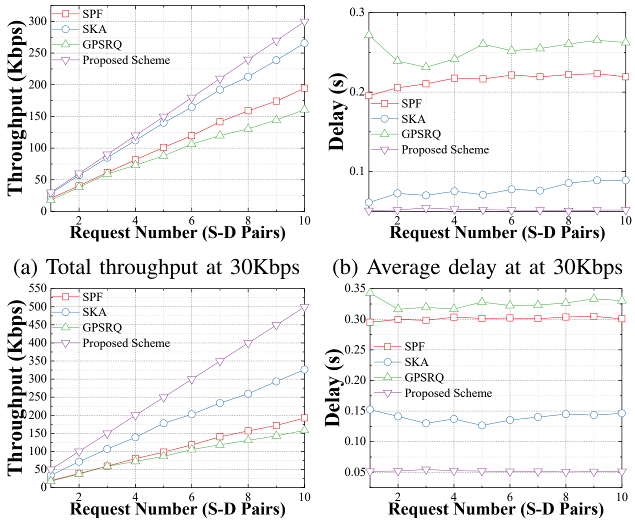
### B. Simulation Results

As shown in Fig. 4, we conduct simulations in a single-request scenario to evaluate the end-to-end key relaying throughput and delay under varying request key rates. As shown in Fig. 4a, our scheme is superior to the baselines in terms of throughput. The multi-path routing in our scheme effectively aggregates the available resources of all links to support the requests, thus satisfying rate-demand QoS requirements better. In contrast, the SPF scheme exhibits a noticeable throughput bottleneck due to its insensitivity to dynamic changes in path resources. The SKA scheme achieves relatively high throughput by periodically updating resource information across the QKD network. However, overlooking the specific requirements of requests limits its performance at a higher request rate. The GPSRQ scheme suffers from frequent rerouting due to a lack of global resource information. As shown in Fig. 4b, our scheme achieves low and stable delays because of the multi-path routing. In contrast, the three baseline schemes experience increasing delays, as they are unable to effectively prevent link congestion.

As shown in Fig. 5, we conduct simulations in multi-request scenarios to further demonstrate the effectiveness of our scheme. The results in Fig. 5a and Fig. 5b show that our scheme maintains its advantages in throughput and delay when the request rate is 30Kbps. This result shows that our scheme is effective when serving multiple requests. As shown in Fig. 5c and Fig. 5d, when the network experiences higher overhead at a request rate of 50Kbps, all four schemes encounter throughput bottlenecks, accompanied by noticeable increases in delay. Nevertheless, our scheme demonstrates superior adaptability by optimizing resource utilization through on-demand detection and refined modeling, thereby meeting the QoS requirements of multiple concurrent requests. Furthermore, the admission control mechanism prevents excessive requests from overloading the network, mitigates link congestion, and helps maintain relatively stable delays.

## VI. Conclusion

This paper focuses on the rate-demand QoS requirements generated by real-time end-to-end communications secured by QKD networks. Thus, we proposed an on-demand routing




(a) Total throughput at 30Kbps (b) Average delay at at 30Kbps

(c) Total throughput at 50Kbps (d) Average delay at at 50Kbps

Fig. 5: Performance comparison results for multi requests.

scheme with QoS provisioning to enhance the end-to-end key relaying rate. By introducing a mechanism of on-demand probing, our scheme gathers global information about the network within a single probe RTT, enabling it to adapt effectively to the dynamic and limited network resources with a minimum cost. Meanwhile, to maximize the service capability of the network, we introduce a maximum flow model and use it to guide the multi-path routing, which not only expands service capability but also incorporates admission control constraints to prevent network congestion. Numerous simulations demonstrate the effectiveness of our routing scheme. It significantly improves the delay and throughput performance for applications with QoS requirements.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